

Reconstructing, Cleaning, and Measuring the Coverage of a Decade of Colorado’s Public Risk-Limiting Audit Data

Abstract

Colorado has an unprecedented record of transparency: a mandatory, statewide risk-limiting audit after every election since 2017, spanning ten years and fourteen election instances. That record has been presented only as a fragmented set of separately published, yearly audit-center exports – three structurally distinct formats – that require substantial reconciliation and cleaning before they support efficient analysis across elections. We reconstruct, clean, and normalize this entire record into a single relational dataset spanning 6,522 contests, verify it against an independent ground truth by reconstructing the state’s own pseudo-random ballot selections from public inputs alone (an exact match for 97.9% of checkable targeted contests), and report the resulting structural findings: 93.9% of contests receive at least some independently examined-ballot evidence, and audit boards’ physical reads disagreed with the voting system’s own interpretation for just 0.067% of 965,254 examined contest-ballot comparisons – a rate dominated by human process error rather than any tabulation flaw. The normalized dataset, analysis code, and a notebook reproducing every result in this paper are available in an open-source, independently reproducible companion repository, laying the groundwork for the further statistical and policy work this reconstruction newly makes possible.

1 Introduction

Colorado has run a mandatory, statewide risk-limiting audit after every election since November 2017 — ten years and fourteen election instances of production data, published by the state as a matter of course. Despite this, no public dataset spans that entire record in a single, normalized form, and no

public analysis has asked the most basic structural questions about it.

We unlock the study of this groundbreaking data by reconstructing, cleaning, and normalizing our archive of CORLA’s exported data into a single relational dataset, and report four findings. First, the raw data itself required substantial cleaning before it could be normalized: three distinct export formats across the survey period, contest identities that sometimes fragment into multiple, differently-named rows within a single election, and unintentional remapping of ballot identifiers caused by common spreadsheet software’s own date-autodetection misreading a hyphenated identifier as a calendar date before the file was published. Second, the resulting dataset can be independently trusted: reconstructing the state’s own pseudo-random ballot selections from public inputs alone reproduces its official selection record exactly for 97.9% of checkable targeted contests, with the small remainder traced almost entirely to specific, well-defined gaps in the available source data rather than any flaw in the reconstruction.

Third, Colorado’s audits produce valuable election integrity evidence relevant to both the formally targeted contests (which achieve their risk limits), and also the other contests which are “opportunistically” audited. Every paper ballot¹ in the sample drawn for a targeted contest’s audit is also interpreted by humans for every other contest present, and compared to the corresponding original voting system record. Fourth, discrepancies between the human interpretations and the system interpreta-

¹A voter’s ballot may consist of more than one sheet of paper; the voting system and the audit both work at the level of an individual sheet, and multi-sheet ballots are interpreted and audited sheet by sheet rather than as one unit. We follow common usage and refer to a ballot-sheet simply as a “ballot” throughout, except where the distinction matters.

tions are rare and mostly due to human error. In the available data, we find a discrepancy rate of 0.067% across 965,254 contest-ballot comparisons, roughly two-thirds of which trace to the audit board retrieving the wrong physical ballot rather than any disagreement about how a given ballot should be read. This reconstruction also surfaces specific gaps in the historical public record — including material once available on the state’s own online audit data site that is no longer, and that this project continues trying to recover — which more systematic archiving of that valuable public resource would help prevent going forward.

We give a brief background on Colorado’s audit design in §2, then describe the data-cleaning methodology (§3), the coverage findings themselves (§4), the verification against an independent ground truth (§5), what the audits actually found when examining ballots (§6), and directions for future work this reconstruction makes possible (§7).

2 Background

Risk-limiting audits provide statistically grounded, software-independent evidence that a reported election outcome matches what the paper ballots actually show [13], rather than merely re-running the same tabulation software again [5] — evidence, rather than trust placed in any single tabulation system, being the foundation the whole approach is built on. Colorado piloted the method years before adopting it statewide: a 2010 audit in one county is credited as the first risk-limiting audit conducted outside California, followed by further county-level pilots in 2011 and a 2013 ballot-level comparison audit of three contests in one county, funded in part by a \$230,000 U.S. Election Assistance Commission grant supporting pilots in five counties from 2011 to 2014, and a further round of mock audits in three more counties in 2015. Risk-limiting audits have since been endorsed by the National Academies of Sciences, Engineering, and Medicine, first in *Securing the Vote* (2018) and again in *Moving to Evidence-Based Elections* (2023) [8, 9]. Colorado has conducted a mandatory, statewide risk-limiting audit after every general, primary, and coordinated election, and before certification, since November 2017, required by Colorado Revised Statutes §1-7-515 [14] and implemented under Secretary of State Rule 25 [2]. Within two weeks after election day, the Secretary of State convenes a public meeting to establish a random seed — a decimal string at least twenty digits long, generated by rolling dice. That seed drives a publicly verifiable pseudo-random

number algorithm to select ballots to be audited, with the aid of a purpose-built comparison-audit tool [1]: a Java web application we refer to throughout as “CORLA”, which county audit boards use to record the audit’s results. CORLA’s role is deliberately narrow: it receives as input ballot manifests and cast-vote-record files (CVRs) uploaded by each county, identifies which paper ballots must be examined, and records each examined ballot’s marks. It then calculates risk levels for the targeted contests, and if necessary, expands the sample size and iterates the process until the risk limit is satisfied.

CORLA’s sampling method is a direct implementation of the standard Kaplan-Markov comparison-audit approach used across many RLA implementations in the United States [11, 6] — one of the two main RLA families alongside ballot-polling audits such as BRAVO [7], which Colorado’s implementation has not used since 2019. For each targeted contest — one whose margin the Secretary of State has selected to formally drive that audit round’s sampling — CORLA hashes the public seed together with an incrementing counter using SHA-256, reduces the result modulo the size of the relevant ballot population, and repeats until enough distinct ballots are drawn to reach a target sample size computed from that contest’s own margin and risk limit. A contest may be targeted at the county level (its own county’s full ballot population is the draw domain), at a multi-county level (the domain is the union of several counties’ manifests, concatenated in a fixed order), or at the statewide level (every county’s manifest, concatenated) — the multi-county case identified as a needed capability, and largely absent from Colorado’s audit software, as recently as 2018 [4].

Not every contest on the ballot is targeted this way. A single physical ballot typically carries dozens of contests — a targeted statewide race, a handful of county-wide offices, and often several much smaller special- district or municipal contests with far narrower margins of their own. When a ballot is drawn to examine its targeted contest, every other contest that ballot also carries gets examined as a byproduct, whether or not that other contest was ever independently selected. We call a contest audited this way — one that receives examined ballots purely as a side effect of some other contest’s targeted draw — opportunistically covered throughout. The distinction between targeted and opportunistically covered contests, and the much smaller set of contests that receive no examined ballots under either mechanism, is the central subject of §4.

We draw on our own practice of periodically down-

loading CORLA’s exported data across every election instance from 2017 through the first half of 2026 — fourteen election instances in all, spanning the variety of export formats described in §3, and reflecting a point-in-time archive rather than a continuous capture of every revision a file underwent (§3.1).

3 Data cleaning and normalization

Colorado has published machine-readable risk-limiting audit (RLA) data for every statewide and coordinated election since 2017 — fourteen election instances spanning ten years. No single, uniform dataset spans that period, however: the state’s audit software and its export conventions changed repeatedly, sometimes within a single election cycle, and the raw exports themselves carry data-quality issues of several distinct kinds. This section describes those changes and issues and how they have been normalized into a uniform schema with mappings back to the original data.

Note also that San Juan County hand-counts its ballots, and does not produce cast-vote records at all. It has not participated in the CVR-based comparison audit this paper’s data comes from. San Juan’s ballots are excluded from every count in this paper for that reason.

3.1 Fourteen elections, three export eras

The audit software’s own export format falls into three structurally distinct eras. The earliest, used for the 2017 coordinated election, the 2018 primary, and the first three rounds of the 2018 general election, is a raw per-cast-vote-record “ballot list” — one row per examined ballot (scanner, batch, and record identifiers, an imprinted ballot ID, and a binary examined/not-examined flag, but no recorded contest choice), with no contest-level parameter file and no standardized per-contest comparison export at all. The 2018 general election’s own final round is a partial exception: it does carry a contest-level parameter file with aggregate discrepancy-type counts, and a per-ballot file recording the audit board’s own read of each examined ballot — but that per-ballot file has no column recording the voting system’s own choice for the same ballot, so it supports aggregate counts but not a genuine two-sided, ballot-by-ballot comparison the way every later year’s export does. The great majority of the fourteen elections, from 2019 onward, use a “contest CSV” era: a per-contest parameter file (`contest.csv` and its several later renamings), a per-contest, per-jurisdiction mapping file, and a genuine two-sided CVR-vs-audit-board

comparison file. Starting with the 2025 CORLA support for auditing Instant-Runoff Voting (IRV) via a different statistical framework (RAIRE-style IRV assertions), rather than Kaplan-Markov comparison auditing, another minor column-name/order change happened.

Even within the “contest CSV” era, filenames and directory conventions vary far more than the underlying column structure does. The per-contest parameter file alone appears under a variety of distinct filenames across the surveyed elections (`contest.csv`, `ContestList.csv`, `ContestsListRound1.csv`), and the comparison file under four (`contestComparison.csv`, `contest_comparison.csv`, `contest_detail.csv`, `CVRtoAuditBoardInterpretationComparison.csv`) — but most of that churn is cosmetic: of the seven file types this schema relies on, five keep exactly the same columns across every election that has them, despite being renamed three to five times apiece. The comparison file itself only changes shape twice in ten years, not once per rename: the 2018 general election’s version lacks a column for the voting system’s own choice entirely (the two-sided-comparison gap already noted above), and the 2022 primary is the first to add a trailing `audit_reason` column, which every comparison file since has carried. Round-numbered directories are named `round_1/round_2/round_3` in earlier years and `round1/round2/round3` (no underscore) in later ones; two elections publish a single, unnumbered `finalReports` directory instead.

To reconcile this we built an explicit, per-election layout table — one small record per election instance naming which filename, directory, and column conventions that instance actually uses. This allows records in the final database rows to reference the exact data they are based on.

A separate limitation, orthogonal to format variation, is that this reconstruction draws on an archive built by repeatedly downloading CORLA’s published exports over time, not a continuous capture of every version of every file. Some files — ballot manifests in particular — can be revised more than once during an active audit cycle as errors are found and corrected, and a snapshot taken between revisions is not guaranteed to be the first version, the last version, or the corrected one. §3.4 documents a concrete case where exactly this pattern combined with a limitation in our own download process to leave a manifest mislabeling uncorrected in the archive well after it was fixed at the source.

3.2 Contest identity: fragmentation and name-variant resolution

A second source of complexity is contest identity itself. The only current way to set the sample size in each county via CORLA is to target a contest and use the global risk limit chosen for all contests targeted in the election. But there is often not a suitable county-level contest available. So sometimes the department of state splits a statewide contest into uses the vote counts within an individual county for a statewide contest to calculate a margin .

CORLA sometimes reports what is, substantively, one contest as several separate rows in the per-contest parameter file: one "bare" row for the contest as a whole, plus one or more county-suffixed rows when that contest's ballot pool in some particular county was too small to support an adequate sample there on its own, forcing a county-specific sub-sample. This pattern — found in five of the eleven checkable election instances — appears under a variety of naming conventions across those five ("**<Contest>** (**<County>**)", "**<Contest>** - **<County>** County", and "**<Contest>** - **<County>**").

The distinction matters beyond bookkeeping. Each county-suffixed fragment carries its own `audit_reason` and its own record of whether that fragment's sub-sample reached its risk limit — and that per-fragment status must never be read as if it were the true, whole contest's status. An analysis that queries a fragment's own "risk limit achieved" field without first resolving it back to its parent contest will systematically over- or under-count how many real contests were adequately audited, because a fragment succeeding or failing says nothing directly about the other fragments of the same true contest. The normalized schema described in §3.5 resolves this by construction: every raw parameter-file row is retained (so no figure is ever un-traceable to its literal source row), but a separate consolidation pass — run once, at import time, for every election — groups fragments back to one true-contest record per real contest, and every downstream table reports only on actual contests.

3.3 An unintentional, fully recoverable remapping of ballot identifiers

The most striking single data-cleaning problem solved in this reconstruction was an unintentional remapping of ballot identifiers, confined to specific export files, that turned out to be entirely recoverable once its mechanism was understood. Col-

orado's ballot- identifier convention is a hyphen-separated triple, `tabulator-batch-position` (for example, `4-5-69`). In eight of the fourteen election instances, a substantial minority of these identifiers instead appear in the comparison export as `M/D/YYYY`-formatted dates — for example, `4/5/1969` in place of `4-5-69`.

The mechanism is demonstrable, not merely plausible. Common spreadsheet software's short-date autodetection reads a hyphenated three-number string as a month-day-year date whenever the first number is a valid month (12 or less) and the second a valid day (31 or less), converts it to an internal date value, and re-emits it in slash-separated form on save, applying a standard two-digit-year century window along the way — an ordinary, well-known spreadsheet behavior, not anything particular to elections data. One of the earliest-era files in the dataset happens to carry the true tabulator, batch, and record numbers in separate columns alongside the remapped identifier in the same row, which makes the mechanism directly verifiable rather than inferred: reversing the transformation on every affected row in that file reproduces the other three columns' values exactly, with zero exceptions across dozens of rows. Because the remapping only triggers when both the tabulator and batch numbers happen to fall within valid date ranges, its incidence varies sharply by county — from under 15% to over 75% of rows in different counties of the same election — purely as a function of each county's own tabulator- and batch-numbering scheme.

Across the full survey (roughly 1,300 candidate files scanned), the remapping appears in exactly the elections using one of three older comparison-file names, and is entirely absent, across several hundred thousand examined ballots, starting with the election that first adopted the current comparison-file name — consistent with an incidental change somewhere in the export or publication pipeline at that point, though nothing in the exported data itself can establish exactly what changed or when. That is not a question this paper attempts to resolve; the more useful fact for this paper's purposes is that the remapping is fully reversible and, once corrected, recovers every one of these identifiers exactly.

The practical value of finding this is straightforward: every remapped identifier would otherwise fail exact-string comparison against correctly-formatted values elsewhere in the same dataset (the ballot manifest, the state's own record of which identifiers were selected for audit), which is what first drew attention to it during selection- reproduction checks. The fix is narrow and reversible — detect the date-

shaped pattern within the known remapping window and invert it, leaving every already-correctly-formatted identifier (the overwhelming majority) untouched — and was verified against the one file where ground truth is independently recoverable before being applied across the full dataset. With it applied, every one of these ballot identifiers is recovered exactly, adding a meaningful share of previously-unmatchable records back into the usable evidence base described in §5.

3.4 Manifest export: mislabeling, gaps, and recovery

Ballot manifests are deliberately independent of the voting system by rule, not merely by incidental county practice: “the county must maintain an accurate ballot manifest in a form approved by the Secretary of State and independent of the voting system” [2]. That independence is what makes the manifest a genuine, separate check on the tabulator’s own count rather than an echo of it — but it also means each county’s own manifest-producing process, not a single state-mandated format, determines the file this paper’s reconstruction has to parse.

Each county’s manifest totals reach this reconstruction through two channels: a standalone per-county manifest export, and a per-batch manifest-vs-CVR comparison file every county also files, which happens to carry the same underlying manifest totals alongside a parallel, CVR-derived count for the same batches. Both describe the same county-reported data; the second is simply a convenient secondary route to the same manifest totals when the standalone export is unavailable.

A separate category of gap affects the ballot manifests themselves. Comparing manifest files byte-for-byte across counties surfaced an instance in 2025 when we pulled the auditcenter at a point in time when the exported Montrose manifest was actually a copy of the Montezuma manifest. The actual sampling data reveals that the audit was working from distinct manifests, since Montrose’s sampled ballots are from tabulator 103, which does not show up in Montezuma’s manifest. So this was clearly an issue with the manual construction of the auditcenter itself, and our scraping software at the time skipped files that had already been downloaded, and wouldn’t catch manual fixes to the export files in auditcenter that came later.

One full election instance was found to have no per-county manifest files published at all; a second was missing manifests for six of sixty-three counties. In both cases, the per-batch manifest-vs-CVR file

described above allowed those gaps to be closed with totals that match the state’s own published aggregate figures exactly, applied county by county rather than as an all-or-nothing fallback, after an earlier all-or-nothing version of the recovery was itself found to be incomplete against affected files that had already been recovered from independent web archive captures.

3.5 Normalized schema for TODO

The result of resolving §§3.1–3.4 is a TODO , with a single relational schema spanning all fourteen election instances, built around one election dimension that every other table references, an explicit true-contest/raw-fragment split that enforces the distinction in §3.2 structurally, and per-row provenance (the literal source filename, and for several tables the full original raw row) retained alongside every derived field, so that any figure quoted from the normalized data remains traceable back to its literal origin. Three election instances prior to 2019 that use the earliest “ballot list” export format populate only the manifest-level tables, rather than forcing a fit into a contest-level schema those exports were never structured to support; one further instance’s IRV audit is scoped out of the risk-bearing tables for the same reason. Nothing in the schema silently drops a known gap — every exclusion in this section is recorded explicitly, as a property of a specific election instance, rather than left for a downstream query to rediscover by absence.

4 Categories of Contests

Colorado’s mandatory RLA program is often described as auditing “the election”, but it targets only a small number of contests by name to achieve the risk limit (§2), and does not report any risk levels for the opportunistic contests.

4.1 Method

Every contest in the normalized dataset (§3) was classified, per election, into one of three categories, based solely on whether and how it appears in CORLA’s own examined-ballot comparison export: directly targeted (the contest itself was named as a formal sampling target), opportunistically named (the contest was never itself targeted, but at least one examined ballot’s comparison record names it — i.e., some ballot drawn for another contest’s draw also happened to carry this one), or no comparison record (the contest’s name never appears in any

examined- ballot comparison record for that election). Each contest was additionally classified by how many counties its own ballot population spans: a single county, a proper subset of the state’s counties (“partial multi-county”), or every county in the election (“statewide”). The three election instances using the earliest export era (§3.1) never populate a per-contest parameter file at all, only manifest-level data, and so cannot be classified by this method — they are omitted from every column below, rather than counted as zero.

One classification exception is known and worth naming directly rather than left as a silent error: “City of Fort Collins Mayor” (2025), an Instant-Runoff Voting contest, is counted below as opportunistically named because this method reads only the state’s main coordinated-election export. A separate, IRV-specific export for that same election (§3.1) shows it was in fact formally targeted and completed its own audit under a RAIRE-based risk-limiting framework, with real, non-zero sampling. That export’s IRV-specific risk methodology is out of scope for this paper (§3.1), but its coverage status should not be — this one contest is misclassified in Table 1, and is named here rather than silently left wrong.

4.2 Results

Across the full survey (Table 1), 6,122 of 6,522 contests (93.9%) have at least one examined-ballot comparison record named in CORLA’s own export: 477 (7.3%) by direct targeting, and 5,645 (86.6%) purely as a byproduct of some other contest’s draw. The remaining 400 contests (6.1%) have no name-matched comparison record in the export for that election. Directly targeted contests are consistently a small minority of the total — never more than about 10%, and often closer to 4% in the larger general elections, where the number of small down-ballot contests grows faster than the number of contests formally selected to drive sampling.

By county-span, the overwhelming majority of contests — 5,223 of 6,522, or 80.1% of the classifiable total — are single-county contests, entirely contained within one county’s own ballot population. A substantial minority (1,215, 18.6%) span a proper subset of the state’s counties without being truly statewide; genuinely statewide contests are comparatively rare (84, 1.3%), appearing only in general elections and the occasional coordinated or primary election with a statewide ballot measure.

4.3 What “no comparison record” does and does not mean

The 400 contests with no name-matched comparison record are not, thereby, contests the audit process skipped. A ballot examined for one contest’s targeted draw is still physically read and interpreted in its entirety by the audit board; it is only when none of the specific ballots that happened to be drawn for other reasons also carried a particular low-prevalence contest that the export ends up with zero rows naming that contest by chance, not by design. This is most common for narrow- prevalence contests — small special districts whose ballot population is a tiny fraction of their county’s full manifest — where the probability that any given examined ballot happens to carry that specific contest can be low even when substantial auditing activity occurred broadly across the same county. Whether, and how precisely, a meaningful statistical conclusion can be drawn for these contests from the sampling activity that did occur nearby is a separate methodological question to be addressed in ongoing work (§7).

5 Dataset integrity verification

The coverage figures in §4 are only as trustworthy as the underlying dataset. Before drawing any conclusion from it, the reconstruction was checked against an independent, external ground truth: does the state’s audit software’s own published selection record match what an outside party, working only from public inputs, can independently recompute?

5.1 Method

For every formally targeted contest — one where the audit software drew a sample directly by name, rather than a contest that merely happened to appear on ballots drawn for another reason — the underlying pseudo-random draw sequence was reconstructed from two pieces of public data: the publicly announced random seed, and the relevant county ballot manifest(s). The audit software’s own selection algorithm, confirmed by direct reading of its published source code [1], is straightforward: hash the seed together with a counter using SHA-256, reduce the result modulo the size of the relevant ballot population, and repeat for each successive counter value until enough distinct ballots are drawn. Reconstructing this sequence independently and comparing the resulting set of ballot identifiers against the audit software’s own official record of which ballots it selected — a separate file the software itself

Election	Total contests	Directly targeted	Opportunistically named	No comparison record	Single-county	Partial multi-county	Statewide
2019	576	52	498	26	457	117	2
2020 (general)	652	60	520	72	510	142	0
2020 (presidential)	2	2	0	0	0	0	2
2020 (primary)	372	34	304	34	282	87	3
2021	666	54	588	24	537	127	2
2022 (general)	961	43	871	47	801	135	25
2022 (primary)	651	38	604	9	543	96	12
2023	661	43	542	76	511	148	2
2024	721	63	597	61	555	142	24
2025	640	62	541	37	515	124	1
2026	620	26	580	14	512	97	11
Total	6,522	477	5,645	400	5,223	1,215	84

Table 1: Contest coverage classification by election, and by county span.

publishes, distinct from the seed and manifest inputs — is the strongest available check: it tests whether the selection itself matches, not merely whether every selected ballot later received a logged comparison record, which is a separate and noisier question addressed by the discrepancy-typing check in §5.3.

5.2 Results

Across the eleven election instances where the necessary files exist (the earliest three instances predate the file formats this check requires, per §3.1), 732 of 748 checkable targeted contests — 97.9% — reproduce the audit software’s own official selection exactly: zero ballots missing, zero extra, for every contest where the check succeeded. Results by election appear in Table 2.

”Blocked” entries are cases where a known, independently-detected manifest gap in the underlying source data (the same kind of gap described in §3.4) prevents any reproduction attempt from running at all — these are counted separately from actual mismatches, and each is a specific, named, previously-documented gap, not a silent skip.

5.3 What the mismatches turned out to be

The 16 remaining discrepancies are informative in their own right: this verification effort is itself a useful way to find data-quality issues, not only a correctness check. Several of the mismatches found in earlier passes of this same reproduction sweep were the direct trigger for identifying the identifier remapping described in §3.3 — ballot identifiers that failed exact-string comparison turned out, on investigation, to have been unintentionally reformatted in the state’s own export rather than genuinely mismatched selections. One further mismatch, a 2025 statewide contest whose domain construction depended on Montrose County’s manifest, is now resolved by the same fix described in §3.4: once Montrose’s true manifest is used instead of its mislabeled

duplicate of Montezuma’s, that contest’s selection reproduces exactly. Of the mismatches remaining after both fixes, most trace to real quirks on the audit software’s own side rather than to the reproduction methodology: one contest’s parameter file carries a stale field left over from an earlier audit round while its companion selection file correctly reflects a later escalation; another contest is administratively classified as spanning the entire state while its actual reproduced ballot identifiers show every hallmark of a genuinely small, few-county domain; several contests share a single already-characterized class of manifest-total gap not yet traced to the specific county responsible. Nine contests’ mismatches remain individually uninvestigated as of this writing, listed explicitly as open items rather than silently omitted.

As a secondary, independent validation channel, a typed-discrepancy check — comparing the count of each Kaplan-Markov discrepancy category (two-vote and one-vote overstatements and understatements, o2/o1/u2/u1; §6.2 reports what these mean and finds substantively) independently derived from the normalized per-ballot comparison data against the audit software’s own reported counts for the same categories, contest by contest — matches exactly for every one of the 477 formally targeted contests where both counts exist (Table 3), a check that exercises a different part of the pipeline (per-ballot comparison parsing rather than PRNG reconstruction) and corroborates the same conclusion from an independent angle.

5.4 What this establishes

A 97.9% exact-match rate against an independently reconstructed ground truth, with every remaining discrepancy individually named and, in most cases, already traced to a specific, understood cause on the source data’s own side, is the basis for treating the normalized dataset as reliable enough to support the structural coverage analysis in §4. It does not, on its

Election	County-level contests	Statewide contests
2019	63/63	1/1
2020 (general)	61/63	0/0 (blocked)
2020 (presidential)	0/0 (blocked)	0/0 (blocked)
2020 (primary)	88/88	1/1
2021	62/64	0/0 (blocked)
2022 (primary)	78/80	2/2
2022 (general)	62/63	0/6 (blocked)
2023	62/63	1/1
2024	62/63	2/2
2025	72/73	2/2
2026	112/112	1/1

Table 2: Selection-reproduction results by election.

Source	o2	o1	u1	u2	Total
CORLA’s own reported count	18	8	12	8	46
Independently derived	18	8	12	8	46

Table 3: Typed-discrepancy validation across just the 477 formally targeted contests: independently derived counts against CORLA’s own reported counts, matching exactly.

own, validate any particular risk-value calculation methodology for contests that were never directly, formally targeted — that is a separate question, addressed in future work (§7).

6 Discrepancies

A key question for both the public and those designing auditing methods is: how often do discrepancies of various types occur, and why? Colorado’s data provides new insights for this question, based on how often the audit board’s physical read of a ballot disagreed with the voting system’s own interpretation, what kind of disagreement it was, and why they might have occurred.

6.1 Method

CORLA’s comparison export records, for every examined ballot-contest pair, both the voting system’s own recorded choice and the audit board’s own hand interpretation; a genuine discrepancy exists exactly when the two disagree in a way that isn’t itself an artifact of the export or a software display quirk (a small number of purely cosmetic candidate-name-formatting mismatches found in one election’s export were confirmed as such and excluded, both here and from CORLA’s own reported counts for that election). After each election, the Department of State also publishes a post-audit discrepancy report, after asking the counties to provide ballot images for each ballot that had a reported discrepancy.

That report assigns a reason category based on their estimation of the most likely explanation for each discrepancy, for example, “Wrong Ballot,” “Audit Board Error,” or “Ambiguous Voter Intent”.

Every genuine discrepancy was classified two ways. First, by the Department of State categorization (when we had a copy of it), and second, by its Kaplan-Markov type — a two-vote or one-vote overstatement or understatement (conventionally $o2/o1/u2/u1$) relative to the specific winner/loser pair the audit’s risk calculation tracks for that contest, or no type at all when the mismatch doesn’t touch that tracked pair (most commonly because the contest is uncontested, so no loser exists to assert against).

These two classifications answer different questions — one is a process explanation, the other feeds into the risk limit calculations. §6.3 shows how they relate. Throughout, “examined” denotes a contest-ballot comparison row, not a distinct physical ballot; a single ballot contributes one row per contest it carries (§2), so the denominator below is substantially larger than the number of physical ballots actually pulled.

6.2 Overall rate

Across every election with ballot-level comparison data (2018 general plus every “contest CSV”-era election, ten instances in all — §3.1), 644 genuine discrepancies were found across 965,254 examined contest-ballot comparisons, an overall rate of 0.067%

— broken down by Kaplan-Markov type and by election in Table 4.

The rate varies by roughly a factor of twenty across elections, from a low of 0.0073% (2018 general) and 0.0076% (2026 primary) to a high of 0.151% (2024 general) and 0.111% (2020 general) — the two largest general elections in the survey, where both the absolute count and the denominator are largest. One instance (2020 presidential, a small two-contest election) shows zero discrepancies of any kind out of 155 examined rows. No secular trend across the ten-year window is apparent; the two general elections have consistently higher rates than the primaries and coordinated elections surrounding them, more consistent with general elections’ higher overall ballot volume and complexity than with any change in audit-process quality over time.

6.3 What causes discrepancies

For the eight elections where we captured the Department of State’s discrepancy report (2020 primary through 2026 primary), 811 discrepancies were attributed to a specific process cause. A single cause dominates: “Wrong Ballot” — the audit board physically retrieved a different ballot than the one the system selected — accounts for 532 of 811 (65.6%), more than three times the next-largest category, “Audit Board Error” (171, 21.1%, a case where the correct ballot was retrieved but its interpretation was mis-recorded). Every other category is comparatively rare: “Voting System Limitation” (an uncorrected undervote the system cannot resolve, 41, 5.1%), “Adjudication Error” (a tabulation-judge misjudgment during canvass, 38, 4.7%), “Ambiguous Voter Intent” (12, and a further 4 recorded under a near- identical, differently-worded category, together 2.0%), and five further categories each appearing in only one or two elections, together under 1% of the total.

Part of Wrong Ballot’s dominance by row count, rather than by underlying incident, is a structural artifact of what “examined” means here (§6.1): a single misretrieved ballot produces one discrepant comparison row for every contest that ballot carries, so one wrong-ballot mistake on a long ballot can register as a dozen or more discrepant rows, while a single Audit Board Error or Adjudication Error typically misjudges just the one contest under review. The reason-category counts above are therefore counts of discrepant rows attributable to each cause, not counts of independent human or process mistakes — Wrong Ballot’s true share of underly-

ing incidents, as opposed to rows, is not established here.

The two classification systems interact informatively. Of 307 two-vote discrepancies (the more severe category, a complete swap between a winner and loser), Wrong Ballot accounts for 222 (72.3%) and Audit Board Error for 83 (27.0%) — together over 99%, essentially the only two causes capable of producing a full name-for-name swap, since both mechanisms substitute one specific candidate’s choices for another’s wholesale. One-vote discrepancies (324 total, a lesser swing toward or away from “neither”) are less lopsided: Wrong Ballot still leads (196, 60.5%), but Adjudication Error’s share jumps to 10.5% (34) from its essentially negligible 0.3% share of two-vote cases — consistent with a tabulation judge’s misjudgment on a hard-to-read mark typically nudging a reading toward or away from non-selection, rather than flipping it to the opposing candidate outright. “Voting System Limitation” never produces a two-vote discrepancy at all (0 of 41), and over half of its rows (24 of 41) register no Kaplan-Markov type whatsoever; tracing those 24 directly shows 21 are contests with only one candidate ever recorded anywhere in the vote totals — mechanically uncontested, so no winner/loser pair can exist for the classifier to assign a type against, regardless of what the ballot itself shows.

The comparison-audit literature sets a rough expectation for how one-vote and two-vote counts should relate, specifically for overstatements: a one-vote overstatement is an ordinary byproduct of how voters mark paper ballots — “occasional one-vote misstatements are expected, owing to the vagaries of how voters mark their ballots: from time to time the system will interpret a light mark as an undervote or a hesitation mark as an overvote” — while a two-vote overstatement, a full swap from one candidate to another, “should be quite rare,” and “generally... indicates a programming error..., fraud, or other serious problem” [6, 12]. This dataset’s own o1 and o2 counts are close in magnitude rather than o2 being negligible by comparison (155 and 153 overall, Table 4) — consistent with our finding that most genuine two-vote discrepancies trace to “Wrong Ballot” or “Audit Board Error”. These are process failures the cited literature’s “programming error, fraud, or other serious problem” framing did not anticipate as the typical cause. Finding the right ballot, especially when the unique “imprinted id” is not physically imprinted on the ballot, requires counting thru the batches of ballots by hand, and an error compounds, as noted above.

Election	o2	o1	u1	u2	Total	Examined	Rate
2018 general	6	3	0	3	12	164,796	0.0073%
2020 (primary)	16	4	4	16	40	61,546	0.0650%
2020 (general)	59	68	71	62	260	234,333	0.1110%
2020 (presidential)	0	0	0	0	0	155	0.0000%
2021	5	11	12	5	33	60,597	0.0545%
2022 (primary)	6	8	8	8	30	118,827	0.0252%
2023	5	3	9	2	19	35,486	0.0535%
2024	47	52	65	61	225	149,272	0.1507%
2025	4	6	4	4	18	47,706	0.0377%
2026 (primary)	5	0	0	2	7	92,536	0.0076%
Overall	153	155	173	163	644	965,254	0.0667%
Overall rate	0.0159%	0.0161%	0.0179%	0.0169%	—	—	0.0667%

Table 4: Two- and one-vote overstatement (o2/o1) and one- and two-vote understatement (u1/u2) discrepancy counts and rates by election. 2019 coordinated and 2022 general are excluded here (source data not further categorized – §6.4); 2017 pilot, 2018 primary, and 2020 presidential’s own predecessor formats have no comparison data at all (§3.1).

6.4 Data-quality gaps in the discrepancy record itself

Consistent with our approach elsewhere (§3, §5), we checked the discrepancy record itself against an independent source rather than taking it at face value, precisely because §6.3’s reason-category findings depend on it: if the underlying row counts were wrong, so would be everything built on top of them. The check did not pass cleanly. We care about this specifically because the affected elections are the same ones cited by name in §6.3, so a reader relying on those numbers deserves to know exactly how far they can be trusted, not a vague disclaimer.

A from-scratch re-derivation of the same underlying Secretary of State reports, performed independently of an earlier published tally of the same eight elections, reproduces four of the eight exactly but comes up 20 rows short across the other four (814 versus 834 total, a 2.4% gap):

One genuine parsing bug was found and fixed in the course of this check — a case-sensitivity error that had silently dropped 16 rows from the 2020 primary’s report, since that one election’s document happens to render “Audit board error” in lowercase where every other year capitalizes it — and is already reflected in the 814 figure above.

Beyond that fix, the remaining 20-row gap across four elections (Table 5) was pursued hard and not resolved. Two independent checks, each using a method structurally unrelated to the original parsing, corroborate the lower, re-derived figures rather than the higher, previously-published ones. First, 2022 primary’s entire source report — both of its rendered pages — was independently re-transcribed

row by row, by a second pass with no access to the parsing code, tallying each category by hand from the rendered text rather than trusting the parser’s own extraction; that from-scratch count also came to 86, matching the parser exactly, category by category, not merely in total. Second, 2021’s count was checked against a completely different data source: a separate method that derives genuine discrepancies directly from raw ballot-level comparison data (CVR versus audit-board interpretation), built independently of this report and without reference to the Secretary of State’s own published document at all, also finds exactly 38 genuine discrepancies for that election — the same lower figure, reached by a method with no shared code or logic with the PDF parser to have propagated a shared bug.

Most strikingly, in three of the four mismatching elections (2020 primary, 2021, 2022 primary), every single reason category is short by exactly one row relative to the originally published total — not concentrated in any particular category, which would be the more typical signature of a single missed or double-counted row. 2020 general breaks this pattern only slightly: four of its five categories are short by exactly one row, while its “Adjudication Error” category matches the published total exactly. This “off by exactly one, spread evenly across categories” signature was actively hunted for a mechanical cause — whether a table row could have split silently across a page boundary, whether two physical rows could have been merged into one extracted text line, whether an alternate spelling of a category name could have been miscounted, whether a column mixup (confusing a “Target Contest” flag with

Election	Independently re-derived	Previously published	Gap
2020 (primary)	119	122	−3
2020 (general)	276	280	−4
2021	38	44	−6
2022 (primary)	86	93	−7
2023	22	22	exact
2024	235	235	exact
2025	22	22	exact
2026 (primary)	16	16	exact
Total	814	834	−20

Table 5: Reconciliation of the independently re-derived discrepancy-report row counts against an earlier published tally of the same eight elections.

an actual “Discrepancy” flag) could explain it — and none of these explains the pattern. The independent row-by-row re-transcription of 2022 primary’s own source pages, described above, found no missing rows there either: the source document itself, read directly, supports 86, not 93.

One concrete, untested hypothesis remains open: 2022 primary’s own report text refers to “the report labeled CVR to audit board interpretation comparison (Round 2)” as “the original version of the report used” — language suggesting the earlier published tally may have been built from a different or earlier version of that same source document, one not retained in this reconstruction’s own archive, rather than from the version available to us. A related possibility is that the earlier tally counted a subtly different thing — for instance, every row flagged as a “Target Contest” rather than only rows specifically flagged as a “Discrepancy.” Neither possibility is confirmed; both remain open questions a direct request to the state for its own original working file could likely resolve.

Given the multiple, methodologically independent corroborations of the lower figures and the inability to locate a further parsing bug despite real effort, we report the independently re-derived counts throughout §6.3 rather than the original published ones. The unresolved 20-row gap is reported here explicitly, not silently absorbed into either number — a specific, well-defined discrepancy, not a vague caveat, and one a direct follow-up with the state could likely resolve.

Coverage of the discrepancy record is also structurally incomplete in ways independent of that reconciliation gap. The three earliest-era election instances (§3.1) carry no ballot-level comparison data of any kind and are entirely absent from every count above. Two further election instances have real, importable comparison data but are deliberately ex-

cluded from the polished figures in §6.2–§6.3 because their underlying source files are documented elsewhere as raw exports never further categorized by the state, with one specifically flagged for a known candidate-ordering software artifact that needs that same qualification attached every time its numbers appear. One further election’s data structurally excludes one county’s ballots entirely, pending resolution of an open question about that county’s relationship to the rest of the dataset. And one instance (2018 general) has only an aggregate, contest-level discrepancy-type count available, with no per-ballot reason-category data of any kind — real numbers, but a coarser measurement than every other election in this section.

7 Preliminary conclusions and future work

A normalized, extensively-cleaned dataset spanning ten years of a state’s production risk-limiting audits facilitates the study of several questions that were previously difficult to approach via the public record.

The next natural goal is calculating risk levels for contests audited opportunistically. Early analysis suggests that a significant fraction of all the contests actually met the risk limit. But a rigorous analysis is complicated. Calculating risk levels for opportunistic contests which cover multiple counties requires a defensible risk-calculation methodology since the sampling rate is not uniform across all the counties. In addition, the current sampling method draws each contest’s audit sample by reducing the seed, a shared per-election random value, against that contest’s own ballot-population size. When two contests’ population sizes share a common factor, this construction means that the samples in the two counties are not independent. Any auditing method that requires independence would need to properly take the correlations into account. Note that this

doesn't affect any of the risk calculations that the audit guarantees for targeted contests.

An excellent way to solve that problem is to adopt Rivest's "consistent sampling" method [10], which derives each ballot's sampling priority from the ballot's own identity rather than a population-size-dependent reduction. It would also be a valuable upgrade for Colorado's audit software since it supports efficient sampling across overlapping contests and supports the many benefits of style-based sampling more generally [3].

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